

Smart AI Medical Report

Generated by Health Navigator

Patient Profile

Name: Mr. Saubhik Bhaumik, Age: , Gender:

Summary of Results

The liver function panel demonstrates all evaluated parameters within the laboratory's reference ranges. The predictive model assesses the profile as within normal limits, indicating no immediate liver pathology is detected.

Detailed Test Explanations

- **SERUM BILIRUBIN (TOTAL):** 0.9 mg/dL (reference 0-1.2). Total bilirubin reflects the balance between hepatic synthesis, conjugation, and excretion. This level is normal, suggesting adequate bilirubin clearance.
- **SERUM BILIRUBIN (DIRECT):** 0.2 mg/dL (reference 0-0.3). Direct (conjugated) bilirubin elevation would signal cholestasis or hepatic cell injury; the value is normal.
- **SERUM BILIRUBIN (INDIRECT):** 0.7 mg/dL (reference 0-1). Indirect (unconjugated) bilirubin rise often indicates hemolysis or impaired hepatic conjugation; the result is normal.
- **SGPT (ALT):** 36 U/L (reference 13-40). ALT is a sensitive marker for hepatocellular injury. The level falls within the normal range, indicating no significant hepatocyte damage.
- **SGOT (AST):** 32 U/L (reference 0-37). AST is less liver specific but, together with ALT, helps gauge hepatocellular injury. This value is normal.
- **SERUM ALKALINE PHOSPHATASE:** Result unavailable. This enzyme can indicate cholestasis or bone disease; absence of data precludes interpretation.
- **SERUM PROTEIN:** 7.2 g/dL (reference 6.4-8.3). Total protein reflects nutritional status and hepatic synthetic capacity; the value is within normal limits.
- **SERUM ALBUMIN:** 4.7 g/dL (reference 3.5-5.2). Albumin is a marker of hepatic synthetic function; the result is normal.
- **GLOBULIN:** 2.5 g/dL (reference 1.8-3.6). Globulin levels remain within the normal range, indicating no significant immunologic or hepatic disturbance.
- **A/G RATIO:** 1.88 (reference 1.1-2.4). The albumin globulin ratio (A:G) falls within normal limits, supporting intact hepatic protein synthesis.

Overall Interpretation

All assessed hepatic enzymes and protein fractions are within laboratory reference ranges. The predictive model flags the profile as normal, underscoring the absence of detectable functional abnormalities or injury. The missing alkaline phosphatase value does not alter this conclusion but warrants eventual measurement for a complete assessment.

Recommendations

- Maintain a balanced diet rich in fruits, vegetables, lean proteins, and whole grains to support hepatic health.
- Limit alcohol intake and avoid hepatotoxic medications or supplements unless prescribed and monitored.
- Continue regular health check-ups; repeat liver function testing annually or sooner if new symptoms (jaundice, abdominal pain, unexplained fatigue) arise.
- If ALP remains unavailable, schedule a repeat test to fully rule out cholestasis or bone pathology.

Disclaimer

This information is for educational purposes only and does not replace professional medical advice, diagnosis, or treatment. Consult your healthcare provider for personalized guidance.